

Code No: R07A1EC10

Set No. 1

I B.Tech Semester Regular Examinations, June 2009
COMPUTER PROGRAMMING AND NUMERICAL METHODS
(Metallurgy & Material Technology)

Time: 3 hours**Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

1. (a) What are the differences between constant and variable?
 (b) Explain about arithmetic operators with examples. [8+8]
2. Give the comparison of switch statement with and nested if statement with examples. [16]
3. (a) Write short notes on pointer arithmetic.
 (b) Explain pointer to pointer with an example. [8+8]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Define ADT . Explain with an example . [16]
6. (a) Find the positive root $x^3 - x = 1$ correct to four decimal places by bisection method.
 (b) Find the parabola of the form $y = ax^2 + bx + c$ passing through the points (0, 0), (1, 1) and (2, 20). [8+8]
7. (a) From the following data, estimate the number of persons having income between
 I. 1000 - 1700
 II. 3500 - 4000

Income	Below 500	500-1000	1000-2000	2000-3000	3000-4000
No. of persons	6000	4250	3600	1500	650

- (b) Find $f(x)$ given the table [8+8]

X	0	1	4	5
f(x)	4	3	24	39

8. (a) Evaluate $\int_0^5 \frac{dx}{4x+5}$ by Simpsons one- third rule and hence find the value of $\log_e 5$ ($n = 10$)
 (b) Given $\frac{dy}{dx} = xy + y^2$, $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2773$, find $y(0.3)$ by Runge-Kutta method of order four and $y(0.4)$ using Milne's predictor-correctormethod.

[8+8]

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Max Marks: 80

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1. What is the structure of while statement? Write a C program to find the sum of series $e = 1 + 1/1! - 1/2! + 1/3! - 1/4! \dots$ [16]
2. Explain about library functions with examples. [16]
3. Write a program to find rank of a matrix using pointers. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Define ADT . Explain with an example . [16]
6. (a) Find the positive root $x^3 - x = 1$ correct to four decimal places by bisection method.
 (b) Find the parabola of the form $y = ax^2 + bx + c$ passing through the points (0, 0), (1, 1) and (2, 20). [8+8]
7. (a) i. Prove that $\delta = E^{1/2} \Delta$.
 ii. Evaluate $(\Delta^2 / E) x^3$.
 (b) Find $f(0.2)$ and $f(5.8)$ by Newton's formula using the data given below [8+8]

X :	0	1	2	3	4	5	6
Y :	176	185	194	203	212	220	220

8. (a) Dividing the range into 10 equal parts, find the approximate value of $\int_0^{\pi} \sin x \, dx$ by (i) Trapezoidal rule (ii) Weddle's rule.
 (b) Given $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$, evaluate $y(1.4)$ by Adams-Bashforth method. [8+8]

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1. What is the purpose of comma and member selection operator in C explain with program? [16]
2. Briefly explain about storage classes. [16]
3. What is the purpose of array of pointers? Illustrate with an example. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Define ADT . Explain with an example . [16]
6. (a) Solve for x from $\cos x - ye^x = 0$ by iteration method.
 (b) Using Newton-Raphson method, Find a positive root of $x^4 - x - 9 = 0$. [8+8]
7. (a) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ$. Find $\sin 52^\circ$, by using Newton's interpolation formula. Estimate the error.
 (b) Find the first and second difference of the polynomial $3x^4 + 8x^3 + 3x^2 - 27x + 9$. [8+8]
8. (a) Find the value of $\log 2^{\frac{1}{3}}$ from $\int_0^1 \frac{x^2}{1+x^3} dx$ by using Simpsons one- third rule with $h=0.25$.
 (b) Using R-K method find $y(.7)$ given $y' = y - x^2$ $y(.6) = 1.7379$. Given $x(0.1) = 2.5257$, $x'(0.1) = -4.820$, $x(0.2) = 2.1040$ and $x'(0.2) = -3.9506$ find $x(0.3)$ and $x'(0.3)$ using 4th order Runge-Kutta method. [8+8]

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Time: 3 hours**Max Marks: 80**

Answer any FIVE Questions
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1. (a) What is the difference between algorithm & flowchart?
 (b) Write briefly about C tokens. [8+8]
2. Give the comparison of switch statement with and nested if statement with examples. [16]
3. Write a program to find rank of a matrix using pointers. [16]
4. Write the definition for Structure and give the general format for declaring and accessing members. [16]
5. Write a program to evaluate the following expression $X = A / B * C$ infix to postfix using stack. [16]
6. (a) Find the positive root $x^3 - x = 1$ correct to four decimal places by bisection method.
 (b) Find the parabola of the form $y = ax^2 + bx + c$ passing through the points (0, 0), (1, 1) and (2, 20). [8+8]
7. (a) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ$. Find $\sin 52^\circ$, by using Newton's interpolation formula. Estimate the error.
 (b) Find the first and second difference of the polynomial $3x^4 + 8x^3 + 3x^2 - 27x + 9$. [8+8]
8. (a) Evaluate $\int_0^3 \frac{dx}{4+x^2}$ by trapezoidal rule and Simpsons 1/3rd rule.
 (b) Given $y'' + xy' + y = 0$, $y'(0) = 0$, $y(0) = 1$ Find the value of $y(0.1)$ using Runge- Kutta method of fourth order. [8+8]
